

ASSIGNMENT 4

GPUs and Cooling System Theory

Q1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

ANS.

❖ **GPU Identification:**

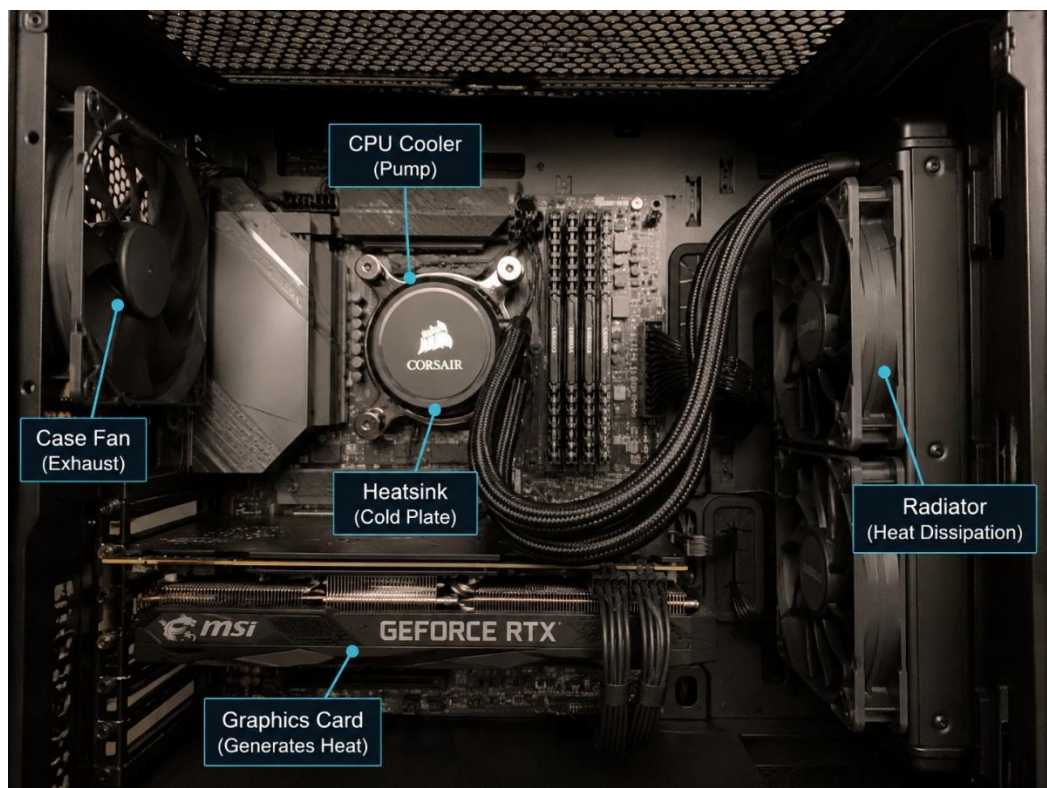
- GPU Brand: AMD
- GPU Model: Radeon 610 Series
- GPU Type: Dedicated Graphics Card (dGPU)

❖ **Explanation (Point to Point):**

- Manufactured by **AMD (Advanced Micro Devices)**.
- **Radeon 610 Series** is an entry-level dedicated graphics card.
- It has its own graphics processor and memory for handling graphics tasks.
- Improves performance for HD video playback, light gaming, and multimedia applications.
- Reduces the graphics workload on the CPU, resulting in smoother overall system performance.

Q2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

ANS.



1.CPU Cooler (Pump):

- **Mounted directly on top of the CPU.**
- **It transfers heat away from the processor and circulates coolant in a liquid cooling system.**

2.Heatsink (Cold Plate):

- **Attached to the CPU beneath the cooler.**
- **It absorbs heat from the CPU and transfers it to the cooler or radiator for dissipation.**

3.Radiator:

- **Located at the front of the PC case.**
- **It removes heat from the liquid coolant with the help of cooling fans.**

4.Case Fan (Exhaust Fan):

- **Installed at the rear of the computer case.**
- **It expels hot air from inside the case, improving airflow and maintaining a lower internal temperature.**

5.Graphics Card (GPU):

- Located in the PCIe slot below the CPU area.
- It generates heat while processing graphics and usually has its own built-in cooling fans and heatsink to keep the GPU at a safe operating temperature.

Q3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

ANS.

Feature	Air Cooling	Liquid Cooling
Cooling Efficiency	Good for normal gaming and everyday use.	Excellent for high-performance gaming, overclocking, and heavy workloads.
Cost	Less expensive and budget-friendly.	More expensive due to pumps, radiators, and tubing.
Maintenance	Requires little maintenance; only occasional dust cleaning.	Requires more maintenance, such as checking coolant and cleaning the radiator.
Noise Level	Can become noisy at high fan speeds.	Usually quieter because the radiator dissipates heat efficiently.
Installation	Easy to install and replace.	More complex to install and requires more space inside the PC.

Q4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

ANS.

Component	Recommended Choice
Processor (CPU)	AMD Ryzen 7 7800X3D or Intel Core i7-14700K
Graphics Card (GPU)	NVIDIA GeForce RTX 4070 Super (or higher)
RAM	32 GB DDR5 (6000 MT/s)
Storage	1 TB NVMe SSD
Cooling System	Liquid Cooling
Power Supply (PSU)	750W 80+ Gold
Motherboard	B650 (AMD) or Z790 (Intel)

- ❖ Choice: Liquid Cooling for the GPU
- I would choose liquid cooling for my GPU because it provides better cooling performance during long gaming sessions and while streaming IPL matches.
 - It keeps the GPU temperature lower, which helps maintain stable performance and prevents overheating.
 - Liquid cooling is also quieter than most air cooling systems under heavy workloads.
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